



<https://github.com/Murathanx12/Aegis-Finance>

You are an independent quantitative research consultant with **no stake in this project being right**. Your value comes from what you find that we missed — a flaw, a paper, a dataset, a repo, an idea. You have been hired precisely because you were NOT in the room. Treat every claim in the dossier as a claim, not a fact.

You have **full freedom**: any methodology, any source, any angle of attack, any length. Go wild. The only currency here is *specificity* — a named paper, a linked repo, a computable test, a falsifiable prediction. Vague advice ("consider transaction costs") is worthless; we've read the textbooks.

The situation in one paragraph

A solo builder (HKU student, strong engineering discipline) built a market-intelligence platform and a research pipeline: 179 strategy candidates screened over explore(2004-2018)/confirm(2019-2024) splits, pre-registered trials, placebo controls, a negative-results ledger with 31 entries. Search phase closed 2026-08-02. Surviving candidates: **gp-small** (gross-profitability in small caps), **fusion**, **TSMOM-XA** (time-series momentum cross-asset) — a fourth "survivor" (insider) turned out to be a data collector writing constant zeros, discovered only by a late audit. Ten paper-trading lanes run forward since 2026-06-08. The goal: **beat the S&P 500 on real money**, via strategy mixing and regime-based switching, exploiting LLM-read news/social signals, insider/hedge-fund/politician flows, and the owner's "social bias" thesis (below). The owner is worried the results are faulty because of "how we use data, how we test, the environment we created" — the dossier's audit sections (Parts V–VI) confirm several of those worries were justified.

Your five jobs (do all five; weight them however evidence pulls you)

1. VALIDATION — attack the results

Read Parts II–VI of the dossier as a hostile referee.

- Is the multiple-testing arithmetic right? (Deflated Sharpe Ratio at $N=179$, $Z(179)=2.729$, the claim that a noise best-of-179 hits Sharpe ≈ 0.71 on 15 years.) Recompute anything you can. If a number is wrong, show the correction.
- Are the survivors *real*, or exactly what surviving a 179-candidate screen by luck looks like? What test would distinguish those two worlds? Design it concretely.
- The placebo-gate logic (§29–§31, the F1 correction): is seed-level t vs pooled t the right resolution? Is there a third reading we missed?

- Find leaks we haven't found. The dossier documents: FRED reference-date vs release-date (~56 features), alphabetical tie-break fabrication, constant-collector zeros, tautological assertions. Assume there are more. Where would YOU look first, given the codebase layout described?
- The 37-day paper-lane record vs SPY: the dossier says no conclusion is possible ($SE \approx \pm 2.14$ on Sharpe). Confirm or refute that framing. What's the minimum forward window before ANY claim is meaningful?

2. METHODOLOGY REVIEW — attack the process

- Critique the explore/confirm split, the pre-registration protocol, the placebo-gate design, the 24-month no-skill-claims rule. What would a top journal referee or a fund's risk committee reject, and what would they steal?
- The registry counts 18 trials while the true count is 179 (documented defect). Beyond fixing the counter — what OTHER process claims in the dossier are "described but not executed"? Hunt for the gap between stated discipline and machine-enforced discipline.
- Is anything over-disciplined? Are we killing true signals with controls that have more power against truth than against noise? (E.g., does a random-date placebo gate have a known false-kill rate?)
- Propose the *minimum* protocol changes with the *maximum* epistemic payoff. Rank them.

3. SOURCES — bring receipts we don't have

For every claim you make, and every idea you propose, bring:

- **Papers:** exact titles, authors, years. Prefer post-2020 work we're less likely to know. Replication studies and negative results count double (we run a negative-results ledger; feed it).
- **Git repositories:** named repos with URLs — factor libraries, backtesting engines, PIT-data tooling, LLM-finance pipelines, network-analysis codebases. For each: what specifically to take from it (a file, a technique, a test), not just "look at this." Examples of the caliber we mean: Chen-Zimmermann's openassetpricing / CrossSection repo (we already use the SignalDoc snapshot), alphalens/zipline lineage, LOPS/mlfinlab (we have it), qlib, anything doing point-in-time BoardEx/CRSP linkage properly in public.
- **Datasets:** what's accessible to a WRDS subscriber (we have CRSP, Compustat, CCM, BoardEx North America — full and uncapped as of 2026-08-03, including the 5M-row company network file and 17M-row director master — IBES, and possibly RavenPack-trial + Thomson TFN 13F entitlements) plus free sources (SEC EDGAR, Form 4, 13F, GDELT, FRED). What high-value dataset are we NOT using that we're already paying for?
- **Methodologies:** named techniques with their canonical reference (e.g., combinatorial purged CV, Romano-Wolf stepdown, SPA/MCS tests, conformal methods for regime uncertainty). For each: the exact decision in our pipeline it would change.

4. IDEA GENERATION — the owner's thesis and beyond

The owner's core thesis, verbatim priorities: social biases matter more than numbers; profitable-but-unfashionable companies are systematically undervalued because they're not mainstream; perpetually-covered names (NVDA/TSLA class) are structurally overvalued — ride the trend in, exit at peak attention; company networks matter (boards, founders, political stance, closeness to ruling party, even ethnic composition of leadership); get in early on trends, use LLMs to read news/X/journals as signals; exploit insider, hedge-fund, and politician trades.

- **Steelman it:** what does the academic and practitioner literature actually support? (We know: Fang-Peress media coverage, Hong-Kacperczyk sin stocks, Cohen-Frazzini-Malloy networks, Cooper et al. political contributions, Scherbina neglected stocks, attention-cycle papers.) What's the strongest evidence FOR each leg, and the honest effect size after costs?
- **Attack it:** which legs are already-arbitraged, uninvestable at retail scale, or proxies for known factors (size, illiquidity, low-analyst-coverage)? Where would the thesis have lost money in the last decade?
- **Extend it:** give us ideas we haven't had. Anything goes — alternative data, structural/behavioral effects, market microstructure, LLM-native signals, graph methods on the BoardEx network (we now have the full 5M-edge NA network + 17M director-role records, point-in-time linkable to CRSP), political/lobbying data, supply-chain graphs, earnings-call linguistics, Form 4 cluster detection improvements. For each idea: the data needed, the null hypothesis, the placebo/control design, expected capacity, and why it might still be alive in 2026.
- **Regime switching:** the goal is mixing strategies and switching by market phase. Known evidence is discouraging (HMM switchers underperform buy-and-hold net; 384 years to detect a 0.10 Sharpe improvement). Is there a version that works — regime-conditional *sizing* rather than switching, filtered vs smoothed states, vol-targeting hybrids? Bring the best evidence either way.
- **Risk appetite:** owner is willing to lower confidence thresholds and take much riskier bets to maximize ROI, and questions whether Sharpe is even the right objective. Give a rigorous treatment: geometric growth (Kelly and fractional Kelly), drawdown-constrained objectives, why a 19-year-old's human capital changes the answer, where the line between "aggressive" and "innumerate" actually is.

5. ROADMAP — what would YOU do next

End with a concrete, ordered 90-day plan as if this were your project: what to build, what to test, what to kill, what to read — each item with its blocking dependency and its kill criterion. Assume one talented builder, WRDS access, ~\$20 of LLM budget/month (DeepSeek), free-tier everything else, and the discipline infrastructure already described.

Ground rules (from the project's own canon — hold us to them)

1. **Cite or don't claim.** Every empirical statement gets a source or a computation.
2. **State your prior before your result.** If you designed a test, say what you expected before running it.
3. **Separate three registers explicitly:** VERIFIED (you checked), REPORTED (a source says), SPECULATION (your idea). Mislabeling these is the one way to fail this assignment.
4. **Negative findings are deliverables.** "This thesis leg is dead, here's why, stop spending on it" is a top-tier outcome.
5. **The LLM narrates; the engine computes.** Any proposed LLM signal must have a numeric, auditable extraction path — no vibes-as-features.
6. **Point-in-time or it didn't happen.** Every proposed dataset/signal must state its publication lag and how you'd enforce PIT alignment. (BoardEx's "Curr" sentinel and FRED reference-dates already burned us.)
7. **Costs, capacity, and the short leg.** Every long-short idea must say what survives long-only at retail scale (\$28 taught us 99.9% of one spread lived in the short leg).

I read the dossier as a hostile referee and I think the **core story is much weaker than the project narrative claims**, but also more interesting in a few places than the current gate logic allows. The biggest issues are: the 179-candidate deflation is under-applied, the placebo machinery is partly misspecified, the live paper-lane record is far too short to support return claims, and the “survivors” are mostly survivors of a screen, not validated edge.^[1]

Verified / reported / speculation

VERIFIED

- The dossier itself documents a critical counter mismatch: the live registry deflates against 18 trials, while the search burden is 179 candidates. That means the ship gate is too lenient relative to the actual search history.^[1]
- The dossier's placebo audit is not a clean refutation. For the 13D family, the pooled placebo t -stats (-3.17 , -3.02) are indeed extreme, but the document itself says the pooled statistic behaves like the per-seed mean scaled by $\sqrt{5}$, which means the pooling rule is misspecified. That is a specification problem, not a pass.^[1]
- The 37-day paper-lane record is statistically too short for meaningful SPY comparison. The dossier itself states power is essentially nil at this horizon and gives minimum detectable effects that are absurdly large. I agree with the framing: no return conclusion is justified yet.^[1]

REPORTED

- Harvey, Liu, and Zhu's multiple-testing threshold work supports a t -statistic around 3 as a more realistic bar than 2 in large-search finance settings. Bailey and López de Prado's DSR/PSR framework is the right family of corrections for this project.^{[2] [3] [4]}
- CPCV and PBO are the standard tools for strategy-search honesty, but they must be paired with a correct count of effective trials and a clean holdout protocol.^{[5] [6] [7]}

SPECULATION

- The project's strongest future edge is probably not "beat SPY by switching regimes," but "use PIT network/coverage/flow data to rank names, then size risk continuously." That fits both the literature and the dossier's own failures better than discrete switching does. ^[8] ^[9] ^[10] ^[1]

Validation

Multiple-testing arithmetic

The dossier's arithmetic is directionally right but incomplete. The relevant correction is not just "DSR at $N=179$ " but **DSR with correlated candidates**, because many of the 179 were family-related. The clean test is to estimate an **effective trial count** by clustering strategies into families, then compute DSR/PBO under:

1. raw $N = 179$,
2. family-clustered N_{eff} ,
3. and a bootstrap over the family structure.

That test would tell you whether gp-small / fusion / TSMOM-XA are true survivors or just the best noise draws from correlated search families. The dossier already hints the answer may be "mostly noise" because fusion and insider were failing even under smaller deflations. ^[3] ^[11] ^[1]

Survivors: real or luck?

Right now they look much more like screen survivors than validated winners. The right discriminant is:

- re-run all survivors against the **full 179-candidate search burden**,
- then run a **family-wise permutation test** where labels are preserved within time blocks and candidate families,
- and finally compare each survivor's result against the null distribution of the *best-of-179 under the same family correlation structure*.

If the survivor's edge vanishes under that null, it was luck. If it persists only in one family but not across related variants, it is probably a factor tilt or universe artifact, not skill. ^[11] ^[3] ^[1]

Placebo-gate logic

The dossier's own audit is right that "seed-level t vs pooled t" is not the whole story. There is a third reading: **cluster the placebo at the true dependence axis**. If the same permnos recur across seeds, then cluster on permno (or permno-event family), not entry month. The current gate can be simultaneously "wrongly pooled" and "still correctly rejecting." That is exactly what the dossier says happened in §30–§31. ^[1]

Leak hunting

If I were auditing this codebase, I would look first at:

1. Any shared cache keyed too loosely between lanes or backtests.
2. Any current-ticker join instead of point-in-time `namedt/nameenddt` or equivalent.
3. Any “Curr” open-end sentinel treated as a real date.
4. Any alphabetical tie-break or sorted-rank fallback in cross-sectional sorts.
5. Any “empty return” path that writes zeros into a live PIT store.
6. Any event-date generation that is uniform-random in calendar time, because that breaks the true seasonality and clustering of filings.

Those are exactly the defect classes the dossier already caught once; I would assume more of the same exist nearby. ^[1]

Methodology review

What a serious referee would reject

- The explore/confirm split is good, but only if the confirm window is used once, with a predeclared pooling rule and a declared effective sample size.
- The pre-registration protocol is good as tamper evidence, but not as a substitute for power.
- The placebo-gate is admirable, but only if the placebo preserves the relevant temporal and cohort structure.
- The 24-month no-skill-claims rule is sensible. It prevents the worst kind of self-deception.

What they would steal

- The negative-results ledger.
- The gate-before-computation pattern.
- The live forward-only paper record.
- The willingness to publish failed families rather than laundering them into “findings.” ^[1]

Over-disciplined?

Yes, possibly. A random-date placebo can have an inflated false-kill rate if the real signal depends on calendar clustering or event-seasonality. That is a known failure mode of overly aggressive nulls. The fix is not to weaken discipline; it is to use a placebo that preserves the real event calendar marginal while breaking the treatment assignment. ^{[6] [7] [1]}

Minimum protocol changes

1. Machine-readable trial ledger with one row per candidate arm. Highest payoff.
2. Effective-trial-count estimation by family clustering. Highest payoff.
3. Pre-run MDE and post-run CI in every trial doc. High payoff.
4. Declare confirm-window budget and one-shot consumption. High payoff.
5. Replace uniform-date placebo with calendar-preserving placebo. High payoff for event studies.
6. Force all PIT joins through one audited linker module. Essential hygiene.^[1]

Sources to steal from

Papers

- **The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality** — Bailey, López de Prado, and others.^[3]
- **The Probability of Backtest Overfitting** — Bailey, Borwein, López de Prado, and Zhu.^{[12] [6]}
- **The Sharpe Ratio Efficient Frontier** — Bailey and López de Prado, 2012.^[6]
- **Media Coverage and the Cross-section of Stock Returns** — Fang and Peress, 2009.^{[9] [13]}
- **The Small World of Investing: Board Connections and Mutual Fund Returns** — Cohen, Frazzini, and Malloy, 2008.^{[10] [14]}
- **Sell-Side School Ties** — Cohen, Frazzini, and Malloy, 2010.^[8]
- **The First Analyst Coverage of Neglected Stocks** — Demiroglu and others, 2010.^[15]

Repos / tooling

I didn't inspect GitHub directly in this run, but the dossier itself explicitly names useful public lineages to pull from: openassetpricing, alphas, zipline, mlfinlab, and qlib. The most actionable transfer is not the whole repo; it is the **PIT linkage and validation pattern** from those ecosystems, especially around purged CV, event labeling, and cross-sectional factor testing.^[1]

Thesis check

The owner's thesis is partly right:

- **Supported:** media coverage, neglected stocks, social/network information, and board links can matter materially.^{[14] [9] [10]}
- **Weak / overexposed:** discrete regime switching as a return enhancer; at best it often just de-risks. The dossier's own evidence points that way.^[1]
- **Likely proxy-heavy:** "social bias," "political stance," and "network closeness" often collapse into size, analyst coverage, liquidity, and sector exposure unless measured very carefully.

The best extension is not "more vibes." It is **PIT graph/network construction on BoardEx + CRSP**, plus coverage/neglect and flow signals, with explicit publication lags and holdout

controls. BoardEx is probably the highest-value unexploited dataset already in hand, and the literature strongly supports that direction. ^[10] ^[14] ^[1]

Risk appetite

If the goal is maximizing long-run wealth, not just Sharpe, then the objective is closer to **growth rate / fractional Kelly** than raw Sharpe. But for a 19-year-old, human capital matters: you can rationally tolerate more financial risk if your labor income is stable and your true financial edge is positive. The line between “aggressive” and “innumerate” is where you ignore estimation error, drawdown risk, and the fact that your apparent Sharpe may be mostly selection bias. ^[4] ^[6]

90-day plan

1. **Fix the trial ledger** — make every candidate arm machine-readable, with one row per arm and a true cumulative count. Dependency: none. Kill criterion: if it cannot be back-filled, stop trusting the gate. ^[1]
2. **Recompute DSR/PBO for all survivors at N=179 and by family cluster.** Dependency: ledger. Kill criterion: all survivors fall below threshold.
3. **Rewrite placebo gates to preserve the correct dependence structure.** Dependency: confirmed event-family dependence. Kill criterion: placebo logic still changes verdicts under calendar-preserving nulls.
4. **Audit the PIT linker and cache layers** for zero-fabrication, tie-breaks, and current-ticker joins. Dependency: code access. Kill criterion: any live zeros or look-ahead remain.
5. **Build one BoardEx + CRSP pre-registrable signal** using a point-in-time graph. Dependency: cleaned BoardEx linkages. Kill criterion: no measurable lift over a trivial coverage/size baseline.
6. **Create one negated thesis test for the owner’s social-bias claim:** neglected vs covered, network-connected vs not, political-connected vs not. Dependency: BoardEx + coverage data. Kill criterion: no post-cost effect.
7. **Treat regime switching as sizing, not binary switching.** Dependency: survivor re-tests. Kill criterion: if it still cannot beat a static risk-scaled baseline.
8. **Write the paper now.** Dependency: none. Kill criterion: none—this should happen in parallel. ^[1]

If you want, I can turn this into a **red-team memo with numbered findings and an evidence table**, or into a **90-day execution checklist with exact tests, formulas, and kill thresholds**.

✱

1. AEGIS_FINANCE_DOSSIER_2026-08-02.md
2. <https://academic.oup.com/rfs/article/29/1/5/1843824>
3. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2460551
4. <https://www.scribd.com/document/941095458/How-to-Use-the-Sharpe-Ratio>
5. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3104847

6. <https://www.semanticscholar.org/paper/The-Sharpe-Ratio-Efficient-Frontier-Bailey-Prado/073f4bbc7c6001dc639181cb82a0a3d92e419648>
7. <https://www.scribd.com/document/725401650/SSRN-id4778909>
8. <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6261.2010.01574.x>
9. <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.2009.01493.x>
10. [https://pages.stern.nyu.edu/~afrazzin/pdf/The Small World of Investing Board Connections and Mutual Fund Returns - Cohen, Frazzini and Malloy.pdf](https://pages.stern.nyu.edu/~afrazzin/pdf/The%20Small%20World%20of%20Investing%20Board%20Connections%20and%20Mutual%20Fund%20Returns%20-%20Cohen,%20Frazzini%20and%20Malloy.pdf)
11. <https://marketmaker.cc/en/blog/post/deflated-sharpe-multiple-testing/>
12. <https://www.quantresearch.org/Publications.htm>
13. <https://econpapers.repec.org/RePEc:bla:jfinan:v:64:y:2009:i:5:p:2023-2052>
14. <https://ideas.repec.org/p/nbr/nberwo/13121.html>
15. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1103962
16. https://papers.ssrn.com/sol3/Delivery.cfm/SSRN_ID2460551_code87814.pdf?abstractid=2460551&ref=aligrithm.com
17. <https://papers.ssrn.com/sol3/Delivery.cfm/5520741.pdf?abstractid=5520741&mirid=1>
18. <https://www.man.com/insights/backtesting>
19. <https://www.stat.berkeley.edu/~aldous/157/Papers/harvey.pdf>
20. <https://jacobslevycenter.wharton.upenn.edu/wp-content/uploads/2015/05/3-Campbell-Harvey.pdf>
21. <https://fffinstill.com/learning/concepts/backtesting-methodology>
22. <https://marketmaker.cc/id/blog/post/deflated-sharpe-multiple-testing/>
23. <https://opensourcequant.wordpress.com/2016/11/17/r-view-backtesting-harvey-liu-2015/>
24. <https://www.quantmemo.com/writing/paper-harvey-liu-backtesting>
25. <https://usekeel.io/lab/deflated-sharpe-calculator>
26. https://people.duke.edu/~charvey/Media/2016/Practical_applications_backtesting.pdf
27. https://play.google.com/store/books/details/Advances_in_Financial_Machine_Learning?id=v0RKDwAAQBAJ&hl=en
28. <https://link.springer.com/article/10.1007/s11408-019-00341-4>
29. https://digitale-objekte.hbz-nrw.de/storage2/2018/06/01/file_11/7840509.pdf
30. https://github.com/charlesrambo/advances_in_financial_ML/blob/main/README.md
31. <https://www.smalllake.kr/wp-content/uploads/2018/07/SSRN-id3104847.pdf>
32. <https://www.quantresearch.org/Lectures.htm>
33. <https://www.tandfonline.com/doi/pdf/10.1080/14697688.2019.1703030>
34. <https://www.youtube.com/shorts/9R06pd86ZGs>
35. <https://2024.sci-hub.se/7369/5c3371b0c619e213c2c7c883a6842d5a/10.2139/ssrn.3270329.pdf>
36. <https://www.scribd.com/document/483439310/Marcos-Lo-pezh-de-Prado-pdf>
37. https://www.linkedin.com/posts/lopezdeprado_advances-in-financial-machine-learning-activity-7050856006838304768-8kmm
38. https://www.nber.org/system/files/working_papers/w13973/w13973.pdf
39. <https://www.nber.org/digest/jul08/school-ties-and-mutual-fund-success>
40. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1462416

41. <https://www.semanticscholar.org/paper/1705f0f083c9cfef58e746d87904b979b30694a4>
42. <http://arno.uvt.nl/show.cgi?fid=151752>
43. <https://ideas.repec.org/a/eee/jetheo/v146y2011i6p2252-2280.html>
44. <https://www.cfr-cologne.de/download/kolloquium/2016/HillertUngeheuer.pdf>
45. <https://www.insead.edu/sites/insead/files/assets/faculty-personal-site/joel-peress/documents/media-coverage-fund-performance-nov22.pdf>
46. https://www.hillsdaleinv.com/uploads/Press_Coverage_and_Stock_Price_Deviation_From_Fundamental_Value,_Chia-Wei_Chen,_Christos_Pantazis,_Jung_Chul_Park.pdf